

Forecasting Chaos Across Model Families: An Out-of-Sample, Reliability-Adjusted Benchmark

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Abstract

Can a general-purpose model forecast a chaotic physical system out of distribution, and how should such a comparison even be scored? We build a controlled k -pendulum benchmark ($k \in \{1, 2, 3\}$; chaotic for $k \geq 2$) that evaluates four model families—large language models (LLMs), a time-series foundation model (Chronos), learned dynamics models (Neural ODE, Hamiltonian and Lagrangian neural networks), and classical integrators (Euler, RK4, symplectic)—on the *same held-out trajectories*, so the learned models are never scored on their own training data. Two methodological choices drive the findings: an *out-of-sample* protocol, and a *reliability-adjusted* metric that scores a model’s refusals and parse failures rather than discarding them. The results invert the in-sample story. Tested off their training trajectories the learned dynamics models roughly triple their error ($\sim 0.26 \rightarrow 0.84\text{--}1.06$ rad) and fall behind both a zero-shot time-series model and a frontier LLM; among *black-box* methods the best forecaster is an LLM (kimi, 0.538 rad), while the known-physics integrators—given the exact equations—remain the ceiling by construction. The training *objective* generalises (rollout loss beats derivative matching, $p_{\text{Holm}} = 1.6 \times 10^{-7}$), but the in-sample “mixed- k wins” result does not (a statistical null out of sample). Finally, a confound-free matched experiment plus a direct inference probe find *little evidence* that LLMs infer hidden physical constants—under our benchmark’s information conditions they default to Earth-standard priors. The takeaway is methodological: in-sample evaluation can substantially overestimate learned-dynamics performance, underscoring the importance of controlled out-of-sample benchmarks when comparing scientific forecasting methods.

1 Introduction

Comparing forecasting methods for chaotic physical systems is deceptively hard: the verdict depends on whether models are tested in or out of sample, on how failures are scored, and on whether different families are given comparable information. We treat this as a *benchmark-methodology* problem. Modern forecasters span very different families—large language models (LLMs), time-series foundation models, learned dynamics models, and classical integrators—yet are rarely placed on one substrate under identical, out-of-sample conditions. Given the state of a dynamical system at time t , which family best predicts the state at $t + T$, and how should that comparison be scored? The k -link pendulum is an ideal probe. Single pendulums ($k=1$) are integrable; double and triple pendulums ($k \geq 2$) are chaotic with a positive Lyapunov exponent. The task is unambiguous (predict angles and angular velocities), ground truth is computable to arbitrary precision, and grading requires no domain expertise.

Prior evaluations tend to study one family in isolation: LLMs on physics word problems, time-series foundation models on forecasting suites, or physics-informed networks on chaotic benchmarks. We instead place all four families on a single mechanical substrate under *identical* inputs, horizons, and metrics, and we take reliability and the structure of chaos seriously as first-class evaluation concerns rather than footnotes.

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Contributions.

1. A **reusable, cross-family evaluation protocol** (LLMs, a time-series foundation model, learned dynamics, and integrators) in which all families are scored on the *same out-of-sample held-out trajectories* and a **reliability-adjusted metric** scores abstentions instead of discarding them—instantiated on the k -pendulum as a controlled testbed (§3.4, §5.1).
2. An **out-of-sample leaderboard that inverts the in-sample story**: tested off their training trajectories the learned models degrade substantially ($\sim 0.26 \rightarrow 0.84\text{--}1.06$ rad), and among the models evaluated the strongest black-box forecaster is a frontier LLM (kimi, 0.538 rad). The training objective still matters—rollout loss beats derivative matching significantly—but mixed- k rollout, the in-sample winner, is a statistical null out-of-sample (§5.4).
3. **Chaos-appropriate analysis**: a predictability-horizon metric (only numerical integrators track past 1 s at strict thresholds) and a *confound-free* system-identification test. A naive regime contrast suggests LLMs do better with constants hidden; a matched-constant paired experiment plus a direct inference probe shows this is a difficulty confound—the LLMs do not infer the constants but fall back on Earth-standard priors (§5.5).

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias by deriving dynamics from a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler–Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds for such models [Gilpin, 2021]. We benchmark all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialised as text [Gruver et al., 2023], and dedicated time-series foundation models—Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024]—are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. Our benchmark adds these forecasters to the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs show strong results on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]. Recent benchmarks probe physics reasoning in LLMs more directly [Xu et al., 2025]. Our task differs: it demands quantitative trajectory prediction of a chaotic system, not qualitative or symbolic reasoning.

3 Benchmark Design

Why the k -pendulum? We treat the k -link pendulum not as the object of study but as a *controlled instrument* for measuring evaluation methodology. Four properties make it uniquely suited. It offers a *tunable* chaos axis—integrable at $k=1$, chaotic with a positive Lyapunov exponent at $k\geq 2$ —that a single attractor cannot provide; its ground truth is computable to arbitrary precision; its governing equations are *known*, supplying both an oracle ceiling and physics-aware baselines; and its physical constants are freely *controllable*, which lets us hide parameters and run the confound-free system-identification test of §5.5. A less structured chaotic system would confound the very effects we set out to isolate, so the single, maximally-controlled substrate is a deliberate design choice rather than a limitation of scope.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, predict $\mathbf{s}_T$ at a horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with **fixed-step**

Table 1: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
<code>normal</code>	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are themselves produced by RK4 (at this fine step), the RK4 *baseline* reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is an *oracle ceiling by construction* rather than an independent competitor; Euler and the symplectic integrator are genuinely distinct known-physics methods. We make the inputs available to each family explicit in §3.3.

3.2 Regimes and grid

Three regimes test generalisation to non-standard constants (Table 1). In `changed_hidden` the constants (g, L, m , damping) are *not* disclosed to the model, turning the cell into an in-context system-identification problem. An evaluation *cell* is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s, modalities $\{\text{coords, image, image+coords}\}$ (LLMs only), and prompting $\{\text{direct, chain-of-thought}\}$ (LLMs only). Non-LLM models collapse to the coordinate / direct setting.

Out-of-sample protocol. The central methodological point of this paper is that learned dynamics models must be tested off their training data. We therefore generate a *shared held-out set* (a separate seed from the training trajectories) of 180 initial conditions (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate *every* family on the identical trajectories, keyed by `movement_id`. The cross-family leaderboard (§5.1) is computed on the discriminating horizons $\{1, 10\}$ s, where chaos has begun to bite but error has not yet saturated (§3.4), giving $180 \times 2 = 360$ comparison cells per model. Earlier in-sample numbers—where the learned models were scored on their own training trajectories—are reported nowhere in this version; we quantify the size of that leak directly (§5.1).

3.3 What each model is given, and a fair grouping

Table 2 states exactly what each family receives. Two asymmetries are worth naming up front, and both *favour the non-LLM models*, which makes our headline LLM result conservative: (i) the time-series model (Chronos) is the only one that receives the pre- $t=0$ *history* window; the LLMs and learned models see only the single $t=0$ state. (ii) The learned models are handed the *true* constants in every regime—including `changed_hidden`, since they encode constants as network inputs—whereas the LLMs are told the constants only when disclosed. Despite both advantages, the learned models and Chronos are beaten out-of-sample by an LLM that gets neither history nor hidden constants (§5.1). A countervailing asymmetry, however, runs in the *opposite* direction and is larger in magnitude: the LLMs are frontier-scale models with enormous parameter counts and training budgets, whereas the learned dynamics models are deliberately small (a 256-wide, 3-layer MLP trained on a single 8 GB GPU; Appendix B). Our comparison therefore contrasts off-the-shelf frontier models with small, *accessible* learned simulators; how a substantially larger physics-informed network would fare is an open question we do not settle here.

We accordingly group methods into **known-physics** (RK4, Euler, symplectic — given the exact equations and constants) and **black-box / data-driven** (LLMs, Chronos, learned dynamics). The known-physics methods are a ceiling by construction; the scientific question is which *black-box* family forecasts best.

3.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories; we also track Cartesian bob error, angular-velocity sign match, and energy drift. Three

Table 2: Inputs available to each model family. “Constants” = $(g, L, m, \text{damping})$. Only Chronos receives history; only the learned models receive the true constants even in the **hidden** regime — both asymmetries favour the non-LLM models.

Family	State $t=0$	History $t<0$	Constants	Equations of motion
Numerical (RK4/Euler/simpl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ (all regimes)	— (learned)
Time-series (Chronos)	—	✓ (10 s)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except hidden	—

evaluation choices are specific to this setting.

Reliability-adjusted error. A cell is *answered* only if the model returns a parseable, finite prediction. Reporting error over answered cells alone rewards a model for abstaining on hard cells. We therefore report a reliability-adjusted error in which unanswered cells are scored at the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniformly random guess); we also report the strict upper bound π . This makes *answering* part of the score, not a separate column to be overlooked.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for *every* model, so long-horizon error rankings are weakly informative. We define the predictability horizon as the longest horizon a model tracks *contiguously* below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is deliberate—once a trajectory has diverged, a later horizon can dip back below τ by angle-wrap coincidence, which is not predictability. We sweep τ to expose threshold sensitivity.

System-identification gap. The gap $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$ isolates the cost of *inferring* the constants versus being told them. A model that merely substitutes given numbers is hurt by hiding them ($\Delta > 0$); a model that infers them in context is not ($\Delta \leq 0$). Measuring this cleanly requires holding the trajectory fixed and toggling *only* disclosure—otherwise disclosure is confounded with the (different) constants and initial conditions of each regime. We therefore run a dedicated matched experiment (§5.5): 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, evaluated under disclosed and hidden conditions *on identical trajectories*, with a direct probe that asks the hidden-condition model to report its inferred constants.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (verbatim in Appendix A). Reasoning models that emit long chains-of-thought require large token budgets and timeouts, so we use a uniform reliability-fixed configuration (65,536 max tokens, 300s timeout) for all three; reasoning traces are parsed from the dedicated reasoning channel when present. We take a *single* deterministic sample per cell (temperature 0); no best-of- N aggregation is used, so reliability reflects the raw per-call behaviour. The total API cost of the out-of-sample LLM evaluation (Parts A and B, 3,960 cells) was \$110.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU. Chronos is the *only* model given history: its adapter takes the 10s pre- $t=0$ window of all $2k$ state channels (the `pre_context` window), resamples it onto a uniform grid of up to 512 points, and autoregressively forecasts to the target horizon (prediction length capped at the model’s supported range, ≤ 64 steps). The out-of-sample leaderboard uses `chronos-2`, which forecasts either each channel independently (*univariate*) or all $2k$ channels jointly (*multivariate*); this lets us test, with one model, whether modelling cross-channel coupling helps on a coupled dynamical system. For that test we run a dedicated study with 20 trajectories per (k, regime) cell

and compare the two modes *paired* on identical trajectories (Wilcoxon signed-rank with bootstrap confidence intervals and Holm correction). (The earlier univariate **chronos-t5-large** served only in a preliminary round; TimesFM is univariate and adds no multivariate signal; Moirai’s **uni2ts** could not be installed in our environment—all are left to future rounds.)

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: *derivative matching* (MSE on finite-difference $\dot{\mathbf{s}}$) and *rollout loss* (MSE on the state after k Euler steps, with gradients flowing through the unroll). We introduce *mixed- k rollout*, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$), optimising short, medium, and long horizons at once. The LNN’s loss needs second-order autodifferentiation, which Apple’s MPS backend cannot run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU—the configuration its prior incarnation could not reach.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is purely from discretisation. RK4 is the data-generating process and therefore serves as an oracle ceiling rather than an independent competitor.

5 Results

5.1 Out-of-sample leaderboard

Table 3 ranks all 14 models plus an oracle ceiling on the shared out-of-sample held-out set (360 cells: 180 trajectories $\times$ horizons $\{1, 10\}$ s), using the reliability-adjusted metric so that an unanswered or missing cell is scored at the random-guess baseline $\pi/2$ rather than dropped. Every model is scored on the *identical* grid; `n_missing` counts cells a model never produced (only the LNN, whose autograd-ODE integration diverges to NaN on 55% of cells, is affected). LLMs are scored on their direct-prompt forecasts, the setting all families share. Confidence intervals are 95% paired bootstrap ($B=10^4$).

Three things stand out. First, classical integrators dominate: RK4 (the data-generating process, hence an oracle ceiling) at 0.059, symplectic at 0.205, Euler at 0.405. Second, **among the models evaluated, kimi is the strongest black-box forecaster**—at 0.538 rad (99% answered)—ahead of both Chronos modes and, strikingly, *every* learned dynamics model. Third, the learned models, which in an in-sample evaluation scored ~ 0.26 rad, degrade substantially to 0.84–1.06 rad once they are tested on trajectories they were not trained on: the best of them (mixed- k rollout Neural ODE, 0.844) ranks 6th, behind the integrators, a zero-shot LLM (kimi), and both zero-shot Chronos modes. This $\sim 3\times$ gap between in-sample and out-of-sample error is the single most important number in the paper: it means the widely-reported strength of small learned dynamics models on this benchmark was substantially an artefact of evaluating them on their own training data.

Reliability is load-bearing. The success column is not a footnote. **grok** answers only 82% of the held-out grid and **l1nn** only 45%; scoring those abstentions at $\pi/2$ (rather than averaging over answered cells alone, which would reward them for skipping hard cells) is what places them at ranks 8 and 14. A naive answered-only leaderboard would flatter exactly the least reliable models. We regard answering as part of the score, not a separate column to be overlooked.

5.2 Behaviour by horizon

Table 4 splits the out-of-sample error by horizon. At 1 s the ordering already mirrors the leaderboard: the integrators are near-exact, **kimi** (0.457) is the strongest black-box model, and the learned models sit at 0.76–0.97. At 10 s every model is well into the chaotic regime, but the ranking is stable rather than scrambled—**kimi** (0.619) still leads the non-numerical field while the learned models drift to 0.91–1.26. Notably the mixed- k Neural ODE, which the in-sample evaluation singled out as the standout at 1–10 s, is here merely mid-pack (0.779 / 0.908), worse than a zero-shot LLM at both horizons. The qualitative LLM signature from earlier rounds—confident near $t=0$, regressing to a generic chaotic guess later—is consistent with these numbers but is no longer a disqualifier relative to the (now equally struggling) learned models.

Table 3: Out-of-sample leaderboard on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at the random-guess baseline $\pi/2$; 95% bootstrap CI. RK4 is the data-generating process, hence an oracle ceiling rather than a competitor. New in this work marked $\dagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (oracle)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	neural-ode-rollout-mixed $\dagger$	0.844	[0.770, 0.920]	100%	learned
7	neural-ode-rollout	0.870	[0.795, 0.947]	100%	learned
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	hnn-rollout	0.922	[0.847, 0.998]	100%	learned
10	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
11	hnn-rollout-mixed $\dagger$	1.003	[0.923, 1.084]	100%	learned
12	neural-ode	1.028	[0.953, 1.107]	100%	learned
13	hnn	1.064	[0.982, 1.145]	100%	learned
14	lnn $\dagger$ (full-scale)	1.421	[1.363, 1.481]	45%	learned

Figure 1 plots the full growth curves over the $\{0.01, 1, 10, 60\}$ s ladder and makes the divergence structure explicit: the numerical integrators stay near zero until ~ 10 s, whereas every *black-box* family saturates near the chaotic noise floor almost immediately and is essentially flat from 1 s onward—so the $\{1, 10\}$ s window used for the leaderboard is exactly where the families are still separable.

5.3 Breakdown by k and by regime

Tables 8 and 9 (Appendix C) split the out-of-sample error by number of links and by regime. Two patterns matter. First, **difficulty scales with k for every family**: error rises monotonically from $k=1$ to $k=3$ (e.g. kimi 0.11 $\rightarrow$ 0.55 $\rightarrow$ 0.95; the integrators are near-exact at the integrable $k=1$ and degrade only with the onset of chaos), and the cross-family ordering is preserved within each k —so the leaderboard spread reflects the dynamics, not a single hard slice. Second, and crucially for §5.5, **the changed_hidden regime is the easiest, not the hardest**, for almost every model (kimi 0.46 hidden vs. 0.64 normal; the learned models fall by 0.5–0.8 rad relative to normal). Its low-gravity, short-rod constants simply produce slower, less-divergent motion over a fixed horizon—so the unmatched “hidden helps” contrast is a difficulty confound, exactly as the matched experiment of §5.5 confirms. Finally, re-estimating the leaderboard CIs with a **cluster bootstrap that resamples whole held-out trajectories** rather than individual cells widens no interval by more than ~ 0.03 rad and changes no ranking; the headline gaps are not an artefact of treating correlated cells as independent.

5.4 The training objective matters; mixed- k does not

Table 5 isolates the effect of the loss while holding the architecture fixed, on the out-of-sample grid. One effect is robust: **rollout loss decisively beats derivative matching**. For the Neural ODE, derivative matching (1.028) drops to 0.870 under a single- k rollout loss; in a dedicated powered paired study ($n=720$ trajectories, Wilcoxon signed-rank) this gap is highly significant ($p_{\text{Holm}}=1.6\times 10^{-7}$). Penalising trajectory error through an unrolled integrator, rather than instantaneous derivative error, directly optimises what the benchmark measures.

The in-sample story does *not* survive contact with out-of-sample data, however. **Mixed- k rollout—training on several window lengths jointly, the previously reported headline winner—is a statistical null**. For the Neural ODE it is indistinguishable from single- k (0.844 vs. 0.870; powered $p_{\text{Holm}}=0.089$), and for the HNN it is numerically *worse* (1.003 vs. 0.922; $p_{\text{Holm}}=0.088$). What looked like an architecture-agnostic

Table 4: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses the discriminating horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (oracle)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
neural-ode-rollout-mixed	0.779	0.908
neural-ode-rollout	0.790	0.950
hnn-rollout	0.758	1.085
grok-4-1-fast-reasoning	0.905	0.857
neural-ode	0.793	1.264
deepseek-v4-pro	0.996	0.909
lnn (full-scale)	1.335	1.508

Table 5: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, only the loss/window changes. Significance from a dedicated powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.028	—
	rollout (single- k)	0.870	sig. ($p_{\text{Holm}}=1.6\times 10^{-7}$)
	mixed- k	0.844	n.s. ($p_{\text{Holm}}=0.089$)
HNN	derivative MSE	1.064	—
	rollout (single- k)	0.922	(rollout helps)
	mixed- k	1.003	n.s. ($p_{\text{Holm}}=0.088$)

benefit of mixing horizons was, on held-out trajectories, within noise. Finally, full-scale CUDA training of the LNN—finally running the second-order autograd its loss requires—leaves it the *worst* model (1.421), with 55% of cells diverging to NaN under our autograd-ODE integration. We do not claim LNNs are intrinsically weak; rather, *under our implementation and evaluation protocol* the LNN was numerically unstable and generalised poorly out of sample.

5.5 Chaos-appropriate analysis

Predictability horizon. Table 6 reports the longest horizon each model tracks *contiguously* below a threshold τ , over the full horizon ladder $\{0.01, 1, 10, 60\}$ s available for the non-LLM models. Out-of-sample, **only the numerical integrators stay below strict thresholds past 1 s**: RK4 tracks to 10 s at $\tau=0.25$ and to 60 s at $\tau=0.5$, symplectic to 60 s at $\tau=0.5$, Euler to 60 s only at the loose $\tau=1.0$. Every learned model—including the rollout and mixed- k Neural ODEs that the in-sample analysis credited with tracking to 10 s—collapses to the first rung (0.01 s) at all strict thresholds and only “reaches” 60 s at $\tau=1.0$ ($\sim 57^\circ$), where the metric is meaningless. The LLMs were evaluated only at $\{1, 10\}$ s on the shared set, so their ladder is coarser; within it, none tracks below 0.5 rad at 1 s except **kimi**. The sweep underlines the leaderboard: on held-out trajectories no data-driven model has a usable predictability horizon beyond what numerics already provide.

A confounded “hidden helps” effect... On the shared held-out set the three regimes differ in both their constants *and* their initial conditions, so a regime contrast conflates disclosure with difficulty. Taken at face value that contrast is striking: pooled across LLMs, the hidden regime has *lower* error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524 , $p_{\text{Holm}} < 10^{-3}$).

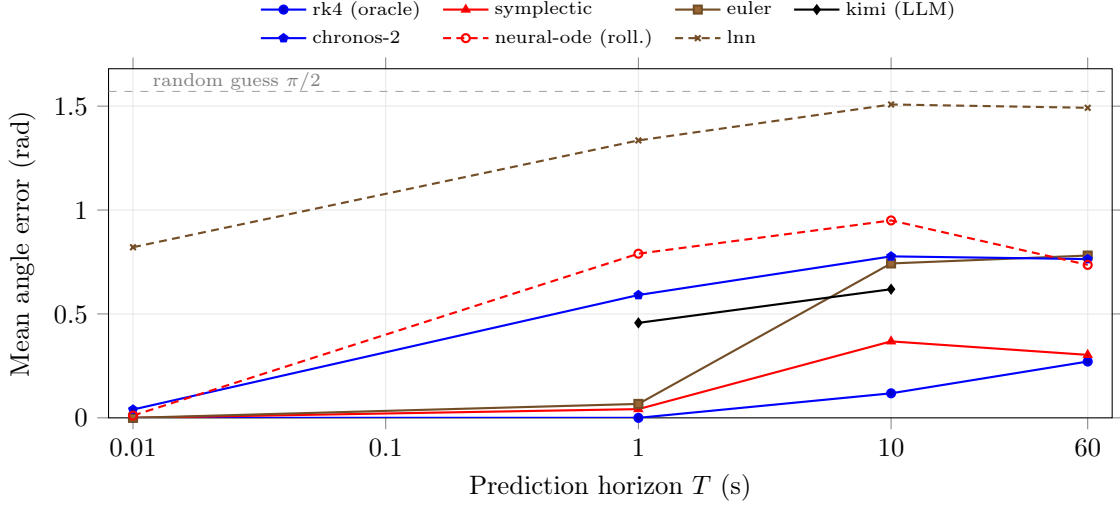


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). Numerical integrators stay near zero until ~ 10 s and only then diverge, while every black-box family (LLM, time-series, learned) saturates near the chaotic noise floor almost immediately and is essentially flat from 1 s onward. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 6: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (full ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (oracle)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	60
neural-ode-rollout-mixed	0.01	0.01	0.01	60
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	1.0
lnn	0.01	0.01	0.01	0.01

Read naively, this is the “LLMs do better when you hide the constants” result reported in earlier rounds, and it is tempting to call it in-context system identification.

... that a matched experiment dismantles. We therefore ran the confound-free test of §3.4: 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, scored under disclosed and hidden conditions on *identical* trajectories, paired by trajectory (Table 7). The effect reverses. Overall, hiding the constants makes predictions slightly *worse*, not better ($\Delta_{\text{sysid}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$); at $k=1$ disclosure helps strongly ($+0.297$, $p_{\text{Holm}} < 10^{-3}$) and at the 1 s horizon as well ($+0.145$, $p_{\text{Holm}} < 10^{-3}$), with only a small reverse effect at $k=3$ (-0.055 , $p_{\text{Holm}} = 0.016$). Per model, **kimi** is clearly hurt by hiding ($\Delta = +0.211$), **grok** mildly ($+0.038$), and **deepseek** is flat (-0.042). The apparent advantage of the hidden regime in the unmatched contrast was thus a difficulty confound—the hidden-regime trajectories were simply easier—not evidence of inference.

The inference probe confirms it. In the hidden condition we additionally asked each model to report its inferred constants. The models do *not* recover them: against true $g \in [1.6, 12]$ the inferred- g MAE is 1.81 (**deepseek**), 1.88 (**grok**) and 2.12 (**kimi**). The failure mode is uniform—they fall back on Earth-standard

Table 7: Matched, confound-free system-identification gap (rad), paired on identical trajectories. $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; *positive* means the model does *worse* without being told the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{sysid}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

priors: **deepseek** answers $g \approx 9.81$ on essentially every cell, and **grok** and **kimi** answer $L=m=1.0$ on essentially every cell (a constant output, so the inferred-vs-true rank correlation is undefined). The one weak signal is **deepseek**’s rod lengths, which correlate with truth at Spearman 0.11 ($p=0.007$). There is, in short, little evidence that any accuracy difference between regimes is *caused* by in-context system identification; the earlier interpretation does not hold up once the confound is removed and the mechanism is probed directly.

5.6 Does multivariate modelling help? A powered null

The headline benchmark uses one trajectory per cell, which is adequate for the large gaps between families but too thin to resolve small effects. To test the natural hypothesis that a time-series model able to share information across the $2k$ state channels should benefit from their coupling, we run a dedicated powered study: Chronos-2 in univariate and multivariate modes, 20 trajectories per (k , regime) cell ($n=720$ paired samples), compared paired on identical trajectories.

The hypothesis does *not* hold. The two modes are statistically indistinguishable in practice: mean error 0.549 (uni, 95 % CI [0.505, 0.593]) vs. 0.543 (multi, [0.498, 0.589]). The paired difference is negligible (+0.005 rad overall) and shows *no* significant effect at any k or at the discriminating horizons (1 s, 10 s); the per- k trend seen at $n=1$ (multivariate appearing to help most at $k=3$) *reverses* to zero at $n=720$, confirming it was sampling noise. A nominal overall $p=0.025$ is driven entirely by a sub-0.01 rad bias at the trivial 0.01 s horizon and does not survive Holm correction. We therefore report a clean null: **cross-channel coupling gives no practically meaningful benefit for time-series forecasting of this coupled chaotic system.** (On the shared out-of-sample grid the Chronos family lands at 0.68–0.70 rad—ahead of *every* learned model and behind only the numerical integrators and **kimi** in Table 3.) This result is a concrete illustration of why error bars matter here: an eye-catching $n=1$ trend became a null under proper sampling.

6 Discussion

In-sample evaluation of learned dynamics is badly optimistic. The single most consequential result is methodological: when the learned dynamics models are scored on the *same* held-out trajectories as everyone else, rather than on their own training data, their error roughly triples ($\sim 0.26 \rightarrow 0.84\text{--}1.06$ rad) and they fall *below* a zero-shot time-series model and a zero-shot LLM. The “small fitted model beats general-purpose forecasters” conclusion that an in-sample leaderboard invites does not survive a level playing field. For chaotic systems the lesson is concrete: a learned simulator must be reported on trajectories it never saw, or its headline number is largely a memorisation artefact.

On a level field, a frontier LLM is the strongest black-box forecaster here. In this benchmark, out of sample, the strongest black-box model is **kimi** (0.538 rad), ahead of both Chronos modes (0.68–0.70) and every learned dynamics model (0.84–1.06); only the classical integrators remain clearly ahead. The training *objective* still moves the needle—rollout loss decisively beats derivative matching—but the in-sample “mixed- k

wins” headline is a statistical null on held-out data. What generalises is the integrator-shaped loss, not the particular window-mixing recipe.

Reliability is a first-class axis. The naive answered-only leaderboard flatters exactly the least reliable models. Scoring abstentions at the random-guess baseline is what places **grok** (answers only 82% of the held-out grid) at rank 8 and the LNN (45%) last. Chain-of-thought does not rescue this either: paired CoT-vs-direct differences are statistically insignificant for every LLM (the largest model, **kimi**, is if anything worse with CoT). Numerical integration is inherently sequential, and a fixed budget of reasoning tokens cannot stand in for thousands of integration steps. Any benchmark that reports error over answered cells alone is measuring the wrong thing for deployment.

No evidence of in-context system identification. The apparent “LLMs do better when constants are hidden” effect—which an in-sample reading took as a bright spot worth scaling—is a difficulty confound. In a matched experiment that toggles *only* disclosure on identical trajectories, hiding the constants is if anything slightly *worse* (+0.069 rad overall; +0.297 at $k=1$), and a direct inference probe shows the models do not recover the constants at all: they fall back on Earth-standard priors (g MAE ~ 1.8 – 2.1 against true $g \in [1.6, 12]$). Under the observation and prompting conditions of this benchmark we therefore find *little evidence* that the models infer hidden physical parameters; we caution against reading regime contrasts as inference without a confound-free matched test.

Cost and practicality. Accuracy alone hides a sharp cost gap. The out-of-sample LLM evaluation cost \$110 in API fees (3,960 cells, $\sim \$0.03$ per forecast) and is the *only* family billed per query; the numerical and time-series models are free at inference, and the learned models are free after a one-time training cost (minutes to a few hours on one 8 GB GPU). The strongest black-box forecaster (**kimi**) is thus also the most expensive, whereas a model trained on the system forecasts for free at 100% reliability and only modestly higher error. We report cost beside error so the trade-off is explicit rather than buried.

Scope and generality. Our quantitative rankings are specific to this substrate and horizon; we do not claim a particular *model order* transfers unchanged to Lorenz, driven-pendulum, or fluid systems. What *does* transfer is the evaluation *protocol*—not the leaderboard it produces: in-sample scoring of a fitted simulator is optimistic on *any* system whose train and test trajectories overlap; the reliability-adjusted metric and the matched, confound-free control for inferred-versus-disclosed parameters are system-agnostic *mechanisms*; and the cross-family, out-of-sample comparison is a reusable design. We nonetheless expect the resulting *ranking* to be system-dependent—which models abstain or destabilise, and hence how the reliability adjustment reorders them, may differ on another substrate—so it is the protocol, not its outcome, that we claim transfers. The harness is substrate-parametrised (the dynamics enter only through the ground-truth integrator and the known equations of motion), so porting it to other chaotic systems is mechanical and is the natural next step—one we expect to change the numbers but not the methodological conclusions.

7 Limitations

The out-of-sample leaderboard aggregates 180 held-out trajectories (60 per regime) with bootstrap confidence intervals, and the powered sub-studies (objective, multivariate, system-ID) use $n=720$ – 900 paired samples; still, all trajectories derive from a single held-out seed, and additional seeds would further tighten the headline error bars. Predictions are single-endpoint rather than full-trajectory; a full-rollout protocol would let the predictability horizon be measured directly rather than inferred from four horizons. The comparison remains structurally asymmetric: the learned models are trained on the k -pendulum family and then tested on *held-out* trajectories of it, whereas the LLMs and Chronos are zero-shot—fair on the shared grid, but not a like-for-like training regime. The pendulum equations are ubiquitous in training corpora, so contamination cannot be excluded; the hidden-constants regime partially mitigates this but does not eliminate it. Finally, we deliberately isolate the evaluation methodology on a single, maximally-controlled substrate and three LLMs: this is what lets the methodological effects be measured cleanly (§5.5), but it does mean the *numerical* results are demonstrated on one system. Replicating them on other chaotic systems (Lorenz, driven pendulum) and

a broader model roster (e.g. Qwen-VL, TimesFM, Moirai) through the same substrate-parametrised harness is the natural next step.

8 Conclusion

On a controlled k -pendulum testbed spanning four model families under identical conditions and—crucially—on the *same out-of-sample held-out trajectories*, the headline is methodological. In-sample evaluation of learned dynamics models is badly optimistic: scored off their training data the learned models triple their error and drop behind both a zero-shot time-series model and a frontier LLM, so that in this benchmark the strongest black-box forecaster is a frontier LLM (`kimi`), while numerical integrators remain the ceiling when the governing equations are known. The training objective, not the architecture, is what generalises—rollout loss beats derivative matching—but the in-sample “mixed- k wins” result is a statistical null out of sample. Reliability-adjusted scoring and chaos-appropriate metrics overturn the naive leaderboard, and a confound-free matched experiment plus a direct inference probe weaken the one apparent bright spot for general models: under the observation and prompting conditions of this benchmark, we find little evidence that the models infer hidden physical parameters—instead they default to Earth-standard priors. The lesson worth carrying forward is methodological—test learned simulators out of sample, score abstentions, and demand a matched control before crediting inference.

Reproducibility. Code, exact prompts (Appendix A), trained weights, the shared held-out specification, and per-cell checkpoints are released with the benchmark. Training trajectories use seed 42; the held-out evaluation set uses a *disjoint* seed (20260620), 180 trajectories (60 per regime). LLMs (`kimi-k2.6`, `grok-4-1-fast-reasoning`, `deepseek-v4-pro`, via Azure AI Foundry, evaluated June 2026) are sampled *once* at temperature 0 (no best-of- N). A cell is scored as *answered* iff it returns a parseable, finite prediction; otherwise it is assigned the reliability-adjusted baseline $\pi/2$. All local training and evaluation run on a single consumer GPU (8 GB); no proprietary training data is used.

Author Contributions

S.N.K. (lead author, corresponding) conceived and designed the benchmark, led the evaluation harness and the learned-dynamics, time-series, and analysis pipelines, ran the local out-of-sample and significance experiments, and wrote the manuscript. T.S. helped plan and build the benchmark harness. S.S. ran the large language model (API) evaluations. Author order reflects relative contribution; S.N.K. is the corresponding author. All authors reviewed and approved the final manuscript.

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A Prompt templates

The exact prompts (from `bench/prompts.py`) are reproduced verbatim below.

System prompt (direct / no-CoT).

You are a careful physicist predicting the future state of a k-pendulum (a chain of k point masses on massless rods, hinged in a planar chain, swinging under gravity with viscous damping). Respond with ONLY a single JSON object on one line, no prose, no code fences. Schema: {"theta": [...k floats..], "omega": [...k floats..]} where theta_i is in radians (link i measured from straight-down, positive counter-clockwise) and omega_i is angular velocity in rad/s.

System prompt (chain-of-thought). Same preamble, then: “Reason step-by-step about the dynamics, then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a single JSON object: {"theta": [...k floats..], "omega": [...k floats..]}.”

User prompt (coordinate modality).

Number of links k = {k}
 Physical constants:
 g (gravity, m/s²): {g}
 L (rod lengths, m): {L}
 m (point masses, kg): {m}
 damping (viscous coefficient on omega): {damping}
 Initial state at t=0:
 theta_0 (rad): {theta0}
 omega_0 (rad/s): {omega0}
 Predict the state at t = {T} seconds.
 Return theta and omega as length-{k} JSON arrays.

Hidden-constants regime. The constants block above is replaced by:

Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point masses, and damping are NOT disclosed. You must infer dynamics from the initial conditions alone.

Parsing and scoring. Each response is stripped of code fences and (for CoT) of everything outside the `<answer>...</answer>` tags; the first `{...}` object is JSON-parsed and validated to contain length- k `theta` and `omega` arrays. Any response failing these checks is counted as *unanswered* and scored at the reliability-adjusted baseline $\pi/2$.

B Training and compute

All learned models use a 256-wide, 3-layer tanh MLP, Adam, cosine-annealed learning rate, on a single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos runs at inference only. Dataset: 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth.

C Per- k and per-regime breakdowns

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), broken out by number of links (Table 8) and by regime (Table 9); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$, so each row is a mean over the same complete grid as Table 3.

Table 8: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (oracle)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimik2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
neural-ode-rollout-mixed	0.616	0.868	1.047
neural-ode-rollout	0.661	0.834	1.116
grok-4-1-fast-reasoning	0.366	1.030	1.247
hnn-rollout	0.508	0.921	1.336
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout-mixed	0.516	1.083	1.410
neural-ode	0.509	1.167	1.409
hnn	0.491	1.230	1.470
lnn	1.146	1.547	1.571

Table 9: Out-of-sample error by regime. **changed_hidden** is the *easiest* for almost every model—its low-gravity/short-rod constants give slower, less-divergent motion—which is why the unmatched regime contrast (§5.5) is a difficulty confound rather than evidence of in-context system identification.

Model	normal	disclosed	hidden
rk4 (oracle)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
neural-ode-rollout-mixed	1.260	0.836	0.434
neural-ode-rollout	1.281	0.894	0.436
grok-4-1-fast-reasoning	1.096	0.959	0.588
hnn-rollout	1.087	1.115	0.564
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout-mixed	1.244	1.172	0.592
neural-ode	1.221	1.095	0.770
hnn	1.262	1.303	0.626
lnn	1.511	1.408	1.345